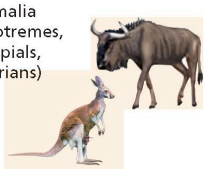
Chapter 34 Review

Campbell Biology, AP Edition, 12e

Chapter 34 Review

Summary of Key Concepts

Identify some of the derived traits in the evolution of vertebrates that enabled many vertebrate species to live on land. **(BIG IDEAS 1 & 4)**

Key Concept	Clade	Description
CONCEPT 34.1 **Chordates have a notochord and a dorsal, hollow nerve cord** *? Describe likely features of the common ancestor of chordates. Explain your answer.*	Cephalochordata (lancelets)	Basal chordates; marine suspension feeders that exhibit four key derived characters of chordates
	Urochordata (tunicates)	Marine suspension feeders; larvae display the derived traits of chordates
CONCEPT 34.2 **Vertebrates are chordates that have a backbone** *? Identify the shared features of early fossil vertebrates.*	Myxini (hagfishes) and Petromyzontida (lampreys)	Jawless aquatic vertebrates with reduced vertebrae; hagfishes have a head with a skull and brain, eyes, and other sensory organs; some lampreys feed by attaching to a live fish and ingesting its blood
CONCEPT 34.3 **Gnathostomes are vertebrates that have jaws** *? How would the appearance of organisms with jaws have altered ecological interactions? Provide supporting evidence.*	Chondrichthyes (sharks, rays, skates, ratfishes)	Aquatic gnathostomes; have a cartilaginous skeleton, a derived trait formed by the reduction of an ancestral mineralized skeleton
	Actinopterygii (ray-finned fishes)	Aquatic gnathostomes; have a bony skeleton and maneuverable fins supported by rays
	Actinistia (coelacanths)	Ancient lineage of aquatic lobe-fins still surviving in Indian Ocean
	Dipnoi (lungfishes)	Freshwater lobe-fins with both lungs and gills; sister group of tetrapods
CONCEPT 34.4 **Tetrapods are gnathostomes that have limbs** *? Which features of amphibians restrict most species to living in aquatic or moist terrestrial habitats?*	Amphibia (salamanders, frogs, caecilians)	Have four limbs descended from modified fins; most have moist skin that functions in gas exchange; many live both in water (as larvae) and on land (as adults)
CONCEPT 34.5 **Amniotes are tetrapods that have a terrestrially adapted egg** *? Explain why birds are considered reptiles.*	Reptilia (tuataras, lizards and snakes, turtles, crocodilians, birds)	One of two groups of living amniotes; have amniotic eggs and rib cage ventilation, key adaptations for life on land
CONCEPT 34.6 **Mammals are amniotes that have hair and produce milk** *? Describe the origin and early evolution of mammals.*	Mammalia (monotremes, marsupials, eutherians) 	Evolved from synapsid ancestors; include egg-laying monotremes (echidnas, platypus); pouched marsupials (such as kangaroos, opossums); and eutherians (placental mammals, such as rodents, primates)

Chordates: notochord; dorsal, hollow nerve cord; pharyngeal slits; post-anal tail

Vertebrates: *Hox* genes duplication, backbone of vertebrae

Cyclostomes: jawless vertebrates

Gnathostomes: hinged jaws, four sets of *Hox* genes

Osteichthyans: bony skeleton

Lobe-fins: muscular fins or limbs

Tetrapods: four limbs, neck, fused pelvic girdle

Amniotes: amniotic egg, rib cage ventilation

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CONCEPT 34.2 Vertebrates are chordates that have a backbone ? Identify the shared features of early fossil vertebrates.	Vertebrates: Hox genes duplication, backbone of vertebrae	Cyclostomes: jawless vertebrates
CONCEPT 34.3 Gnathostomes are vertebrates that have jaws How would the appearance of organisms with jaws have altered ecological interactions? Provide supporting evidence.	Gnathostomes: hinged jaws, four sets of Hox genes	
CONCEPT 34.4 Tetrapods are gnathostomes that have limbs ? Which features of amphibians restrict most species to living in aquatic or moist terrestrial habitats?	Osteichthyans: bony skeleton	Lobe-fins: muscular fins or limbs
CONCEPT 34.5 Amniotes are tetrapods that have a terrestrially adapted egg ? Explain why birds are considered reptiles.	Tetrapods: four limbs, neck, fused pelvic girdle	Amniotes: amniotic egg, rib cage ventilation
CONCEPT 34.6 Mammals are amniotes that have hair and produce milk ? Describe the origin and early evolution of mammals.	Mammalia (monotremes, marsupials, eutherians)	Evolved from synapsid ancestors; include egg-laying monotremes (echidnas, platypus); pouched marsupials (such as kangaroos, opossums); and eutherians (placental mammals, such as rodents, primates)

Concept 34.7: Humans are mammals that have a large brain and bipedal locomotion

- Derived characters of humans include bipedalism and a larger brain and reduced jaw compared with other apes.

- **Hominins**—humans and species that are more closely related to humans than to chimpanzees—originated in Africa 7–8 million years ago. Early hominins had a small brain but probably walked upright.
- The oldest evidence of tool use is 2.5 million years old.
- *Homo ergaster* was the first fully bipedal, large-brained hominin. *Homo erectus* was the first hominin to leave Africa.
- Neanderthals lived in Europe and the Near East from about 350,000 to 28,000 years ago.
- *Homo sapiens* originated in Africa about 195,000 years ago and began to spread to other continents about 180,000 years ago.

Based on fossil evidence, summarize how key hominin features have changed over time.